

# Younsoo Park

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## Education

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### The Pennsylvania State University

University Park, PA

Bachelor of Science in Computer Science Minor in Mathematics Expected Graduation: May 2027 Dean's List

## Research Interests

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Systems Security Network Security Distributed Systems Federated Learning TinyML for IoT Security Adversarial Machine Learning Edge AI Systems

## Research Experience

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### Undergraduate Research Assistant Team Lead

Oct 2025 – Present

*Reliable Federated TinyML Deployment for IoT Security (WIP, LCTES '26)*

Penn State University and Elizabethtown College Supervisors: Dr. Suman Saha and Dr. Peilong Li

Research focus: Federated and adversarially robust TinyML systems for IoT security and resource-constrained edge-device deployment.

Key contributions:

- Designed a federated learning pipeline using FedAvgM, focal loss, and cosine learning-rate scheduling on CIC-IDS2017.
- Improved attack recall from 46.7
- Built a TinyML compression pipeline including BatchNorm folding, structured pruning, knowledge distillation, and INT8 QAT/PTQ.
- Achieved  $12.28\times$  model compression and 74.5
- Evaluated adversarial robustness using PGD attacks under multiple compression configurations.

## Publications and Manuscripts

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**Reliable Federated TinyML Deployment for IoT Security** Work in Progress Target Venue: ACM LCTES 2026

## Industry and Technical Experience

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### SIGNUM (Nittany AI Alliance – PIT-UN Funded)

Aug 2024 – Nov 2025

Co-founder of a \$3,750 PIT-UN funded AI platform for hospital quality analysis and caregiver decision support.

- Built multi-source ETL pipelines integrating CMS, NPES, and Google Places datasets into a DuckDB warehouse.
- Developed hospital-rating prediction models using Markov Transition analysis with confidence intervals.
- Designed and deployed domain-specific RAG systems using FAISS and AWS Bedrock.

- Built secure authentication systems using SQL/JWT authentication and encrypted credential storage.
- Implemented automated credential generation and authenticated onboarding workflows.

## Technical Projects

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**ASME @ Penn State Website** Oct 2025 – Present

- Developed the official ASME Penn State website using React, TypeScript, Vite, and Firebase.
- Designed a multi-tier permission and approval system for organization management.

**Hangukgwan (Family Business Web App)** 2024 – Present

- Developed a multilingual full-stack restaurant web application using React, TypeScript, Node.js/Express, and MongoDB Atlas.
- Implemented JWT authentication, Google Maps API integration, and multilingual i18n support.

## Teaching Experience

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**Lead Teaching Assistant, Math 230 (Multivariable Calculus)** Penn State University Aug 2025 – Present  
Responsibilities include assisting instruction, supporting students during office hours, and helping coordinate course activities.

## Leadership and Service

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**THON Technology Committee** Technology Captain (2027) Apr 2026 – Present

**Penn State Advanced Vehicle Team (AVT)** Project Manager Fall 2026 – Present

**Association for Computing Machinery (ACM)** Web and Design Director Apr 2026 – Present

**Republic of Korea Air Force** Squad Leader Sept 2022 – June 2024

## Relevant Coursework

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### Computer Science

- CMPSC 131: Programming and Computation I
- CMPSC 132: Programming and Computation II
- CMPSC 221: Object Oriented Programming
- CMPSC 311: Introduction to Systems Programming
- CMPSC 360: Discrete Mathematics
- CMPSC 461: Programming Language Concepts
- CMPSC 464: Theory of Computation
- CMPSC 465: Data Structures and Algorithms

## Computer Engineering

- CMPEN 270: Digital Design
- CMPEN 331: Computer Organization and Design

## Awards and Honors

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|---------------------------------------------------------|----------|
| 1st Place, ThinkNeuro LLC Project Competition           | Apr 2026 |
| 1st Place, IEEE Student Chapter Battle Bots Competition | Apr 2026 |
| PIT-UN Funding Award (\$3,750)                          | Apr 2025 |
| 3rd Place, ACM Pathfinder Robotics Challenge            | Apr 2025 |
| Republic of Korea Air Force Commendation Award          | Aug 2023 |

## Technical Skills

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**Programming Languages:** Python, C, C++, Java, JavaScript, TypeScript, SQL

**Frameworks and Tools:** React, Node.js, Express, Flask, MongoDB, Firebase, Git, AWS Bedrock, DuckDB

**Machine Learning:** TensorFlow, PyTorch, FAISS, SHAP, spaCy, Scikit-learn, Flower